

a. Thallus with gemmae cups

b. Male gametophytes bear antheridia

c. Female gametophytes bear archegonia

Figure 23.7 Liverwort, *Marchantia*. **a.** Gemmae can detach and start a new plant. **b.** Antheridia are present in disk-shaped structures. **c.** Archegonia are present in umbrella-shaped structures. (a, c): Ed Reschke/Stone/Getty Images; (b): J. M. Conrader/Science Source

groups of cells that detach from the thallus and can start a new plant. Sexual reproduction depends on disk-headed stalks that bear antheridia, as well as on umbrella-headed stalks that bear archegonia. Following fertilization, tiny sporophytes, composed of a foot, a short stalk, and a capsule, begin growing within archegonia. Wind-blown spores are produced within the capsule.

Hornworts

The **hornwort** gametophyte usually grows as a thin rosette or ribbon-like thallus between 1 and 5 cm in diameter. Although some species of hornworts live on trees, most live in moist, well-shaded areas. They photosynthesize, but also have a symbiotic relationship with cyanobacteria, which, unlike plants, can fix nitrogen from the air.

The small sporophytes of a hornwort resemble tiny green horns rising from a thin gametophyte, usually less than 2 cm in diameter (Fig. 23.8). Like the gametophyte, the sporophyte can photosynthesize, although it has only one chloroplast per cell. Meiosis occurs within the "horn," and spores are released. Alternatively, hornworts can bypass alternation of generations by reproducing asexually through fragmentation.

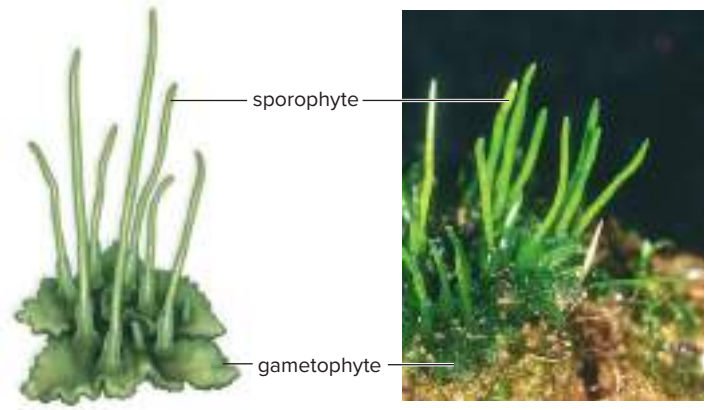


Figure 23.8 Hornwort, *Anthoceros* sp. The "horns" of a hornwort are photosynthetic sporophytes that grow continuously from a base anchored in gametophyte tissue. (photo): Steven P. Lynch

Mosses

Mosses are the largest group of nonvascular plants, with over 12,000 species. There are three distinct groups of mosses: peat mosses, granite mosses, and true mosses. Although most prefer damp, shaded locations in the temperate zone, some survive in deserts, and others inhabit bogs and streams. In forests, they frequently form a mat that covers the ground and rotting logs. In dry environments, they may become shriveled, turn brown, and look completely dead. As soon as it rains, however, the plant becomes green and resumes metabolic activity.

Members of the moss genus *Sphagnum* (peat moss) have great commercial and ecological importance to humans. The cell walls of peat moss have a tremendous ability to absorb water, which is why it is used in gardening to improve the water-holding capacity of the soil, and why it was used historically as wound dressing on battlefields. One percent of Earth's surface is peatlands, where dead *Sphagnum* accumulates but does not decay.

Figure 23.9 describes the life cycle of a typical temperate-zone moss. The gametophyte of mosses begins as an algalike, branching filament of cells, the protonema, which precedes and produces upright, leafy shoots that sprout rhizoids. The shoots bear either antheridia or archegonia. The dependent sporophyte consists of a foot, which is enclosed in female gametophyte tissue; a stalk; and an upper capsule, which contains the sporangium, where spores are produced. A moss sporophyte is always attached to the gametophyte. At first, the sporophyte is green and photosynthetic; at maturity, it is brown and nonphotosynthetic. In some species, the sporangium can produce as many as 50 million spores. The spores disperse the new gametophyte generation.

Check Your Progress 23.2

1. Explain the various methods of bryophyte reproduction.
2. List the characteristics that enabled the bryophytes to successfully colonize land.
3. Explain which portion of the bryophyte life cycle is the most dominant.

ISTUDY